

# Machine Learning for Aquatic Remote Sensing

[As a part of the ISC-SCOR workshop being held at Behrampur University, India]

This two-part training introduces participants to the intersection of data science and hydro-optics. While traditional band-ratio algorithms have served the community well, the complexity of Case-2 waters often requires non-linear solutions. This workshop combines theoretical foundations with practical, hands-on coding exercises, enabling participants to understand, build, and validate Machine Learning (ML) models for water quality retrieval. The focus of this workshop is to move beyond simple implementation and enable participants to build models that are not only accurate but also explainable. Detailed session curriculums described below:

## **Session 1: Introduction to ML in Aquatic Remote Sensing**

This session begins with a comprehensive tour of the current machine learning state of knowledge, establishing taxonomy of algorithms – from Decision Trees (Random Forest, XGBoost) and Support Vector Machines to Deep Learning. We will also understand exactly where empirical algorithms fail and ML excels in optically complex waters. We will then pivot to the concept of “Scientific ML”, distinguishing it from standard data science by emphasizing physics-informed learning that respects constraints (such as non-negative concentrations) and addresses the bias-variance tradeoff in the context of sparse in-situ data. The session will *make visible* the “black box” of AI through an introduction to Explainable AI (XAI) techniques, demonstrating how to use Feature Importance to understand the spectral drivers of a model. We will conclude with a critical look at data ethics and reporting, where participants will learn to navigate the tradeoffs of using log-scale and select robust metrics, going beyond  $R^2$  (like Bias, RMSE, MAPD) to ensure their results are scientifically rigorous.

## **Session 2: Hands-on ML/AI for Aquatic Remote Sensing**

In this hands-on session, participants will transition from theory to practice by building a complete water quality retrieval workflow using Python and Google Colab. The session starts with Exploratory Data Analysis (EDA) to visualize spectral relationships and target distributions, providing the necessary context for training a Random Forest regressor. Participants will actively code the model implementation and practice hyperparameter tuning to improve generalization. A major focus will be placed on rigorous validation and interpretation; participants will generate publication-quality validation plots and utilize SHAP (SHapley Additive exPlanations) values to explain individual model predictions. The workshop culminates in a practical application where users apply their trained models to actual satellite imagery (e.g., PACE or Sentinel-3 or Landsat) to generate and visualize spatial water quality maps.

**Target Audience:** Graduate students, researchers, and professionals in marine science, oceanography, coastal management, and environmental monitoring. This workshop is designed for those who have a grasp of traditional remote sensing concepts but wish to transition to rigorous, data-driven ML approaches.

### **Prerequisites**

- *Remote Sensing:* Basic understanding of aquatic optics ( $R_{rs}$ , IOPs) and satellite data products.
- *Computing:* Familiarity with basic programming concepts (Python or R) is helpful. The hands-on session will utilize Google Colab, so a Google account is required.

### **Learning Outcomes**

By the end of this workshop, participants will be able to:

1. **Differentiate** between standard ML and "Scientific ML," applying principles that ensure physical plausibility.
2. **Train** effective regression models (Random Forest/XGBoost) for retrieving Chl-a or TSM from spectral data.
3. **Implement Explainable AI (XAI)** techniques (such as SHAP values) to interpret model decisions and identify physical drivers.
4. **Critique** model performance using appropriate metrics and visualizations, avoiding common reporting pitfalls.
5. **Navigate** the ethical considerations of using Generative AI tools (LLMs) for scientific coding assistance.

### **Technical Requirements**

#### **Online Platforms (No Installation Required)**

- *Google Account:* To access Google Colab notebooks.
- *GitHub Account* (Optional): To access code repositories and datasets.

**Desktop Software (Optional):** Participants wishing to run code locally post-workshop should have a Python environment (e.g., Anaconda) installed with the following libraries:

- scikit-learn, shap, pandas, numpy, matplotlib, rasterio

For any questions before, through or after the workshop, please contact

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